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Electrodeposition of zero-valent selenium nanoparticles onto glassy carbon electrode from ethaline deep eutectic solvent

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Abstract

Zero-valent trigonal selenium nanoparticles (SeNPs) were potentiostatically formed at room temperature (298 K) on the surface of a glassy carbon electrode (GCE) from SeO_2 dissolved in ethaline, a deep eutectic solvent composed of choline chloride and ethylene glycol. The mechanisms and kinetics of SeNP electrochemical nucleation and growth on GCE were investigated using both potentiodynamic and potentiostatic techniques. From cyclic voltammetry, the equilibrium potential (E_{eq}) of the Se(IV)/Se(0) redox couple, the exchange current density (j_0), and the charge transfer coefficient (α) for the faradaic reaction $Se(IV)_{ethaline} + 4e^-_{GCE/Se(0)} \rightleftarrows Se_{(s)}$ were determined to be E_{eq} = 96 mV vs. Ag QRE, $j_0 = (1.2 \pm 0.1)$ $\mu A \cdot cm^{-2}$, and $\alpha = 0.4 \pm 0.1$. Analysis of the experimental potentiostatic current density transients (j–t plots) recorded at various overpotentials revealed that the SeNP electrodeposition mechanism involves three consecutive processes: (*i*) Adsorption and double-layer charging, (*ii*) Multiple 3D nucleation and diffusion-controlled growth of SeNPs, and (*iii*) Residual water reduction on the growing SeNP surfaces ($2H_2O_{DES} + 2e^-_{(Se)} \rightleftarrows H_{2(g)} + 2OH^-_{(DES)}$). The applied theoretical model enabled deconvolution of the total j–t response into these individual contributions, allowing for extraction of key kinetic parameters, including the SeNP nucleation frequency (A), the number density of active nucleation sites (N_0), and the diffusion coefficient of Se(IV) in ethaline, $D_{Se(IV)}$ (298 K) = $(4.91 \pm 0.04) \times 10^{-9}$ $cm^2 s^{-1}$. Furthermore, from the dependence of A on overpotential, beside j_0 and α, additional thermodynamic parameters were determined: the surface energy (σ), the critical nucleus size (n_c), and the Gibbs free energy for the formation of the critical nucleus ($\Delta G(n_c)$) across the range of applied overpotentials. SEM analysis revealed the formation of smooth, quasi-spherical SeNPs with an average diameter of (118 ± 26) nm, uniformly distributed across the GCE surface. High-resolution XPS spectra recorded in the Se 3d region confirmed the presence of selenium in its zero-valent state. The Raman spectrum displays characteristic signals at 236 and 141 cm^{-1}, indicating that the predominant allotropic form of selenium in the nanoparticles is trigonal Se. Additionally, the optical band gap of the electrodeposited film was estimated using UV–Vis spectroscopy and the Kubelka–Munk method. A band gap of 1.7 eV was obtained, which is consistent with previously reported values for trigonal Se nanoparticles.

Keywords Selenium · Nanoparticles · Electrodeposition · Nucleation and growth · Ethaline

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Introduction

Selenium nanoparticles (SeNPs) possess unique physicochemical properties that make them valuable for a wide range of applications, including biomedical [1, 2], environmental [3, 4], and catalysis-related fields [5, 6]. Typically, less than 100 nm in size, SeNPs can interact with biological systems at the cellular level. In biomedical contexts, SeNPs have been studied for their antioxidant [7] and anti-inflammatory properties [8], as well as their potential in cancer therapy due to their ability to selectively target tumor cells [9].

In the field of energy, SeNPs have been explored for enhancing the performance of solar cells and other energy conversion devices [10]. Additionally, SeNPs have shown promise in the development of chemical, electrochemical, and biosensor devices for the detection of hydrogen peroxide, heavy metals, and glucose [11], these nanoparticles exhibited a distinct spherical morphology, contributing to their increased surface area and enhanced electrocatalytic activity. For constructing SeNP-based electrochemical sensors, it is essential to both synthesize SeNPs and immobilize them onto suitable substrates to create functionalized electrode surfaces. In this context, Kuang et al. [12]. demonstrated the utility of poly(ionic liquid) composites as solid electrolytes in lithium-metal batteries, highlighting the expanding role of ionic liquid–based media in high-performance energy storage applications.

Various methods have been employed for SeNP synthesis, including physical (e.g., gamma irradiation, hydrothermal microwave, pulsed laser ablation, sonochemical), chemical (e.g., chemical reduction), biological (e.g., using plants, microorganisms, biomolecules, and biowastes) [11], and electrochemical (electrodeposition) using both; aqueous [13–16] and non-aqueous media [17–19]. Among these methods, electrochemical electrodeposition offers distinct advantages: it allows direct synthesis of SeNPs on conductive electrode surfaces in a single step, with precise control over nanoparticle size and morphology by tuning the deposition parameters such as applied potential, time, and temperature. Moreover, it is a simple, cost-effective, and scalable method.

Deep eutectic solvents (DESs) [20] have recently emerged as promising media for nanoparticle synthesis via electrochemical methods [21–24]. DESs, typically composed of a quaternary ammonium salt and a hydrogen bond donor, offer desirable properties such as low toxicity, low volatility, high ionic conductivity, and the ability to dissolve a wide variety of compounds.

Despite these advantages, there remains a lack of systematic studies investigating the mechanisms and kinetics of SeNP nucleation and growth during electrodeposition from DESs. Although some works have reported Se [18] or Se-alloy [25–28] electrodeposition from DESs, the underlying electrochemical nucleation/growth mechanisms and detailed characterization of the resulting SeNPs have not been thoroughly explored. Furthermore, SeNP-decorated structures may also be of interest for advanced energy storage applications, since, as shown by Yang et al. [29]., nanocomposites obtained through specific element doping and surface decoration can significantly enhance the electrochemical performance of such materials.

Therefore, in this work, we report for the first time a comprehensive study on the mechanism and kinetics of electrochemical nucleation and growth of SeNPs onto glassy carbon electrode surfaces from SeO_2 dissolved in ethaline (a DES composed of choline chloride and ethylene glycol) at 298 K. Ethaline exhibits a wide electrochemical stability window, which enables the reduction of selenium species (e.g., Se(IV)/Se(0)) without premature solvent decomposition. This feature is crucial for selenium electrodeposition, as its reduction potential may lie outside the stability range of many protic or aqueous electrolytes. Furthermore, ethaline is non-volatile and thermally robust compared with water and many organic solvents, which minimizes solvent loss at elevated temperatures that may be used to optimize deposit quality and reduces safety concerns associated with flammable organic media. This also improves reproducibility over long electrolysis runs. Moreover, due to its viscosity [30] and conductivity, ethaline is one of the few DES that can be used adequately at room temperature. Furthermore, glassy carbon was selected as the working electrode because it provides a chemically inert, electrochemically stable, and highly reproducible platform, enabling the investigation of selenium electrodeposition without interference from substrate-related reactions. In comparison with metallic or oxide-based substrates, GCE allows a clearer interpretation of deposition kinetics, nucleation mechanisms, and film growth behavior in deep eutectic solvents. In addition, GCE is fully compatible with post-deposition characterization such as: SEM/EDS (no substrate interference), Raman spectroscopy and XPS (absence of overlapping metallic core levels). This is especially important for selenium, where oxidation states and bonding environments are often inferred from surface-sensitive techniques.

Experimental

Ethaline DES preparation

Ethaline DES was prepared using a 1:2 molar ratio of choline chloride (ChCl) and ethylene glycol (EG). The appropriate masses of both components were weighed using an analytical balance to form the eutectic solution that was then heated on a hotplate at 70 °C for 4 h under constant stirring to facilitate the formation of the ionic liquid. Once the DES was synthesized, a 50 mM Se(IV) was prepared by dissolving Selenium (IV) oxide (SeO_2 98% purity) in the DES. All reagents used were analytical grade sourced at Sigma-Aldrich. The residual water content of the DES was determined by Karl Fischer coulometric titration using a Titrino Coulometer model 756 from Metrohm® giving less than 0.1%.

Electrochemical measurements

Electrochemical studies were carried out using a Biologic bipotentiostat fitted with EC-Lab software to perform both cyclic voltammetry (CV) and chronoamperometry (CA). The electrochemical cell consisted of a three-electrode system: a GCE was the working electrode (3 mm diameter), a graphite rod (99.997% purity, acquired from Goodfellow advanced materials) as the counter electrode, and a silver wire as the quasi-reference electrode (Ag QRE). It is important to mention that Ag QRE potential, when immersed in ethaline at room temperature, is quite stable (at least for 20 h), as demonstrated by Shen et al., see curve (2) in Fig. 3b [31]. All electrodes were thoroughly cleansed before each experiment to prevent contamination, repeating the cleaning protocol every measurement cycle. The glassy carbon electrode (GCE) surface was polished using alumina powder, thoroughly rinsed with deionized water, and subsequently immersed in deionized water and sonicated for 5 min. The electrode was then rinsed again with deionized water prior to use. The cell temperature was maintained at 298 K.

Electrodeposits characterization

The morphology of the electrodeposited SeNPs over the glassy carbon electrode was characterized with a high-resolution Scanning Electron Microscopy (SEM) JEOL, JSM-6701 F, using secondary electrons and Inca software. after selenium electrodeposition, the working electrode surface was rinsed with deionized water using a wash bottle to remove excess deep eutectic solvent from the surface. The deposit composition was established by means of (EDX) coupled to the SEM and X-ray photoelectron spectroscopy (XPS) using a Escalab 250 Thermo Scientific instrument fitted with an Al filament set at 1486.6 keV driven by the Avantage 5 software for acquisition and processing of data.

Raman spectroscopy was performed using a Renishaw inVia system equipped with a 532 nm green laser operated at 50% power, with 10 accumulations of 5 s each. Finally, UV-Vis diffuse reflectance spectroscopy (UV-Vis DRS) was conducted using a UV-vis spectrophotometer with Agilent Technologies Cary 1G Solids Diffuse Reflectance accessory to determine the band gap of the obtained samples.

Results and discussion

Se(IV) speciation in ethaline is generally expected to involve coordination with oxygen atoms from the ethylene glycol component, rather than the formation of strong Se–Cl bonds, which may dominate in other chloride-rich but less strongly coordinating solvents. However, detailed information on this important aspect of the newly explored Se(IV)–ethaline system is currently lacking. A thorough investigation of the structure of the predominant Se(IV) species in this medium would require advanced analytical techniques, such as X-ray absorption spectroscopy (EXAFS/XANES), UV–Vis spectrophotometry, and nuclear magnetic resonance (NMR) spectroscopy. Such an analysis lies beyond the main scope of the present work.

Potentiodynamic study

Figure 1 shows a cyclic voltammogram (CV) recorded with a glassy carbon electrode (GCE) immersed in an ethaline deep eutectic solvent containing Se(IV) species. The CV exhibits a characteristic crossover loop between the forward and reverse potential scan branches, which is indicative of nucleation and growth processes associated with the

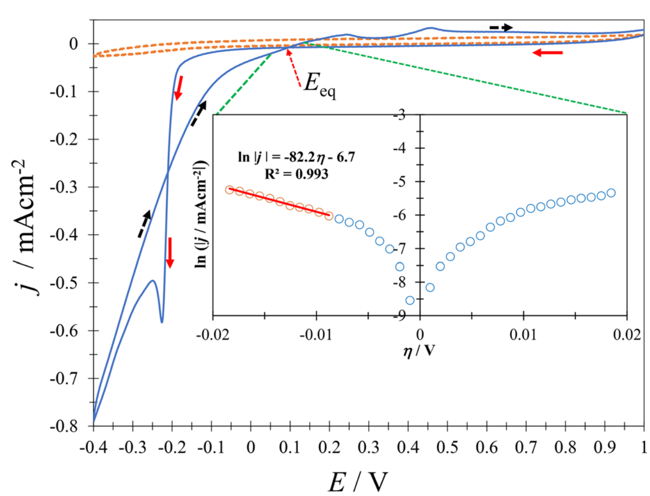

Fig. 1 Experimental CVs obtained in the system: GCE / 50 mM Se(IV) in the ethaline DES (solid blue line) and GCE / DES (broken orange line). In both cases, the temperature was 298 K and the potential sweep started at 0.8 V, in the negative direction, at 2 mV s^{-1}, the arrows indicated the forward (red) and backward (black) potential scan. The inset shows the variation of ln $|j|$ with the overpotential, η, recorded around the equilibrium potential, E_{eq}, of the Se(IV) / Se(0) couple, along with the cathodic branch of the Tafel plot, the points are the experimental values (each current density value is the average of three experiments) and the red line and equation correspond to their linear fit

formation of a solid phase on a substrate of different nature. Since the only solid phase that can be formed under these conditions is zero-valent selenium, Se(0), the electrodeposition process necessarily involves a four-electron transfer. ($Se(IV)_{DES} + 4e^{-}_{\mathbf{GCE}} \rightleftarrows Se_{(s)}$). During the forward (cathodic) potential scan, the current remains negligible until approximately −0.1 V. Beyond this point, a well-defined cathodic peak emerges, corresponding to the reduction of $Se(IV)_{DES}$. In the reverse (anodic) scan, the presence of a pronounced crossover loop [32] is observed, indicating that selenium is electrodeposited onto GCE throughout a nucleation and growth mechanism. On the forward scan, reduction of selenium species begins only after a significant overpotential due to the nucleation energy barrier. Once Se nuclei are formed, they provide energetically favorable sites for further growth. On the reverse scan, deposition continues at potential values more positive than those required during the forward scan. This results in a current crossover, producing a nucleation loop. The equilibrium potential (E_{eq}) for the Se(IV)/Se(0) redox couple, where the deposited phase is in quasi-equilibrium with the electrolyte, corresponds to the crossover potential, at which the forward and reverse scans intersect within the nucleation loop region at zero current density (the rates of deposition and dissolution are equal) [33], represents the thermodynamic equilibrium potential for the Se(IV)/Se(0) couple under the given experimental conditions (solvent, temperature, concentration). It is important to note that the current density at the equilibrium potential, E_{eq}, appears to be negative, whereas it should ideally be close to zero. However, this cathodic (negative) current is also present in the blank electrolyte, as shown by the dashed orange line. Therefore, it can be attributed to electrochemical double-layer charging rather than to a faradaic process. Indeed, if the CV recorded in the blank solution is subtracted from the CV obtained in the presence of Se(IV), the resulting current density at E_{eq} becomes negligible, as expected for a system at equilibrium. Moreover, the low potential scan rate used to record the experimental cyclic voltammograms was deliberately selected to ensure that quasi-equilibrium conditions were achieved, which is required for an accurate determination of the equilibrium potential (E_{eq}) using the potentiodynamic technique and subsequent application of the Tafel equation, as discussed below. From this crossover loop, the experimental E_{eq} of the Se(IV)/Se(0) redox couple was estimated to be +96 mV vs. Ag QRE (see Fig. 1). For potentials more positive than E_{eq}, anodic current densities are detected; however, these are significantly lower than the cathodic currents. This suggests that the oxidation of Se(0) to Se(IV) ($Se_{(s)} \rightleftarrows Se(IV)_{DES} + 4e^{-}_{\mathbf{GCE/Se(0)}}$) is either passivated or kinetically irreversible under the conditions studied. The inset in Fig. 1 presents the corresponding Tafel plot, constructed by plotting ln|j| as a function of the overpotential ($\eta = E - E_{eq}$), obtained from linear sweep voltammograms (j vs. η) recorded during the reverse scan of the CV. This portion of the voltammogram represents selenium reduction and oxidation occurring on/from pre-deposited selenium on the GCE surface, described by the reversible reaction: $Se(IV)_{DES} + 4e^{-}_{\mathbf{GCE/Se(0)}} \rightleftarrows Se_{(s)}$. The cathodic branch of the Tafel plot ($\eta < 0$) was analyzed using the Tafel approximation, see Eq. (1), enabling the estimation of the exchange current density (j_0) and the charge transfer coefficient (α) for the previous electrode reaction. The values obtained were: $\alpha = 0.4 \pm 0.1$ and j_0 = (1.2 ± 0.1) μA·cm^{-2}.

$$\ln|j| = \ln(j_0) - \alpha z F R^{-1} T^{-1} \eta \tag{1}$$

Where j is the current density, F is the Faraday constant, R is the universal gas constant, T is the temperature, z is the number of electrons transferred and η is the overpotential.

Potentiostatic study

Figure 2a presents a series of potentiostatic current density transients (j–t plots) recorded within the overpotential range where Se(0) nucleation and growth occur. All j–t transients were recorded using a single potential step, which in all cases started at +0.6 V vs. Ag QRE, a potential at which no Faradaic current was observed and the GCE surface was free of deposited selenium (see Fig. 1). The potential was then stepped to the selected value required to achieve the overpotential considered in this work. All curves exhibit characteristic features: an initial sharp current drop at short times, followed by a rise to a maximum (j_m) at time t_m. After reaching this peak, the current gradually decreases and eventually approaches a steady-state value. The initial current decay has been commonly attributed to pseudo-capacitive processes, including double-layer charging and adsorption phenomena [34–39].

A key observation, shown in Fig. 2b, is that the experimental j–t transients deviate significantly from those predicted by the classical 3D nucleation models proposed by Scharifker and Hills [40], particularly for t/t_m > 1. This behavior has been previously described by Palomar-Pardavé et al. [41] and attributed to the simultaneous occurrence of multiple processes: 3D nucleation and growth of the metallic phase under diffusion control, and a concurrent faradaic reaction such as residual water reduction on the surface of the growing nuclei. Similar deviations have been reported during the electrochemical nucleation of Ni [42–44], Cr [24, 45], Pd [46–48], Al [23, 49], Fe [50], Au [51], and Nd [21] nanoparticles on glassy carbon electrodes from DES. Interestingly, Bougouma et al. [18] also reported this behavior for the dimensionless current transients recorded

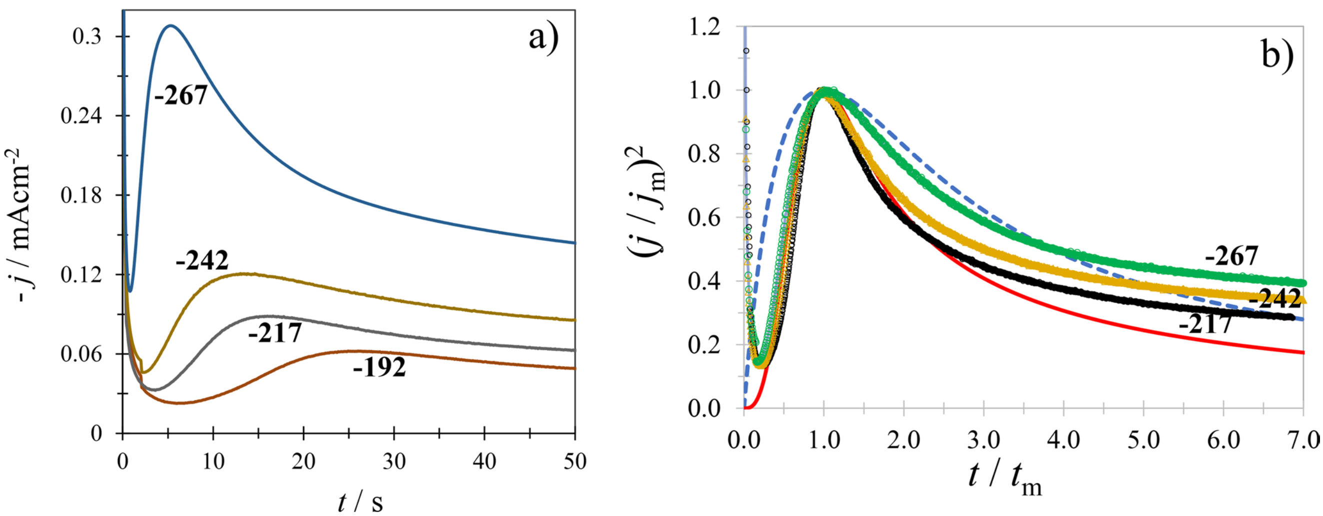

Fig. 2 (**a**) Family of experimental potentiostatic current density transients recorded in the system GCE / 50 mM Se(IV), dissolved in ethaline DES at 298 K at the different overpotentials indicated in the figure, in mV. (**b**) Comparison of the non-dimensional plot of experimental current density transients (points) recorded at different overpotential (see Fig. 2) with the theoretical ones for instantaneous (broken blue curve) and progressive nucleation (red curve) mechanisms [40]

during selenium electrodeposition from reline DES onto a polycrystalline gold electrode (see Fig. 8 in Ref [18]). Moreover, the corresponding *j*–*t* plots exhibit very similar shapes to those obtained in the present work (see Figs. 6 and 7 in Ref [18]).

Moreover, Lai et al. [15] also reported positive deviations from ideal nucleation behavior in the electrodeposition of Se on tin oxide from aqueous H_2SeO_3 + LiCl solution (pH 2.0). In their case, deviations at $t > t_m$ were ascribed to concurrent proton reduction on the surface of the deposited Se film (see Fig. 5 in [15]) however, the experimental *j-t* plots were not analyzed.

Taking these findings into account, the analysis of the *j*–*t* transients in Fig. 2a was conducted using Eq. (2), which assumes that the total current comprises three distinct contributions: *i)* j_{ad} (*t*): associated with adsorption and double-layer charging processes; *ii)* $j_{3D\text{-dc}}$ (*t*): arising from 3D nucleation and diffusion-controlled growth of Se(0) nanoparticles (SeNPs) and j_{WR} (*t*): due to the electrochemical reduction of residual water on the surface of the growing SeNPs, according to the reaction: $2\ H_2O_{DES} + 2e^-{}_{SeNPs} \rightleftarrows H_2(g) + 2OH^-{}_{DES}$.

For the 3D nucleation term, $j_{3D\text{-dc}}$ (*t*), the model developed by Heerman and Tarallo [52, 53], as adapted by Palomar-Pardavé et al. [41], was employed (see Eq. 52 in [41]). The Dawson integral appearing in the Heerman-Tarallo expression was approximated by a polynomial as reported by Arbit et al. [54]. and is shown in Eq. (10).

$$j_{total}(t) = j_{ad}(t) + j_{3D-dc}(t) + j_{WR}(t) \tag{2}$$

With

$$j_{ad}(t) = k_1 exp(-k_2 t) \tag{3}$$

$$j_{3D-dc}(t) = P_1 t^{-1/2} \frac{\Phi}{\phi} \theta(t) \tag{4}$$

$$j_{WR}(t) = P_3 \theta(t) \tag{5}$$

With

$$k_1 = \frac{E}{R_s} \tag{6}$$

$$k_2 = \frac{1}{R_s C} \tag{7}$$

$$\theta(t) = (1 - exp(-P_2 t \varphi(t))) \tag{8}$$

$$\varphi(t) = 1 - \frac{1 - exp(-At)}{At} \tag{9}$$

$$\Phi(t) = 1 - \frac{1 - exp(-At)}{(At)^{\frac{1}{2}}} \int_0^{(At)^{\frac{1}{2}}} exp(\lambda)\, d\lambda \cong 1 - \left[\frac{a + b(At)^{\frac{1}{2}}}{\left(1 - c(At)^{\frac{1}{2}} + d * At\right)(At)^{\frac{1}{2}}} \right] \tag{10}$$

$$P_1 = \frac{ZFD^{\frac{1}{2}}\, C_0}{\pi^{\frac{1}{2}}} \tag{11}$$

$$P_3 = (2\pi)^{\frac{3}{2}}\, D\,(MC_0/\rho)^{\frac{1}{2}}\, N_0 \tag{12}$$

$$P_3 = \left(\frac{2C_0M}{\pi\rho}\right)^{\frac{1}{2}} Z_{WR}Fk_{WR} \tag{13}$$

In these equations R_s is the solution's resistance, C the double layer capacitance, C_0 and D *are* the bulk concentration and diffusion coefficient of Se(IV) respectively, M and ρ are the atomic mass and density of Se respectively, $z_{WR}F$ is the molar charge trasferred during the water reduction, k_{WR} is the rate constant of the water reduction reation on the selenium deposit surface, A is the SeNPs nucleation frequency and N_0 is the number density of active sites for Se nucleation on the electrode surface. The numerical values a=0.051314213, b=0.47910725, c=1.2068142, d=1.185724 correspond to the parameters of polinomial approximation to the Dawson integral.

Therefore, Eq. (2) can be parameterized as follows:

$$j\,(t) = k_1exp(-k_2t) + \left(P_3 + P_1t^{-1/2}\right)\left(\frac{1-\left[\frac{a+b(At)^{\frac{1}{2}}}{\left(1-c(At)^{\frac{1}{2}}+d*At\right)(At)^{\frac{1}{2}}}\right]}{1-\frac{1-exp(-At)}{At}}\right)\left(1-exp(-P_2t(1-\frac{1-exp\,(-At)}{\mathrm{A}t}))\right) \tag{14}$$

Figure 3a compares the experimental current transients with theoretical curves obtained from non-linear fitting of Eq. (14). The corresponding individual current contributions are shown in Fig. 3b, exemplifying the ability of the model to deconvolute the total current into the distinct physical processes occurring simultaneously. Figure 3c presents j_{WR} (t) as a function of time and overpotential, using the best-fit parameters (Table 1) and Eqs. (5), (8), (9), and the empirical P_3 parameter. It is evident that with increasing overpotential, the time to reach the maximum water reduction rate decreases. Furthermore, P_3 varies linearly with η, while both the nucleation frequency, A, and P_2, relates

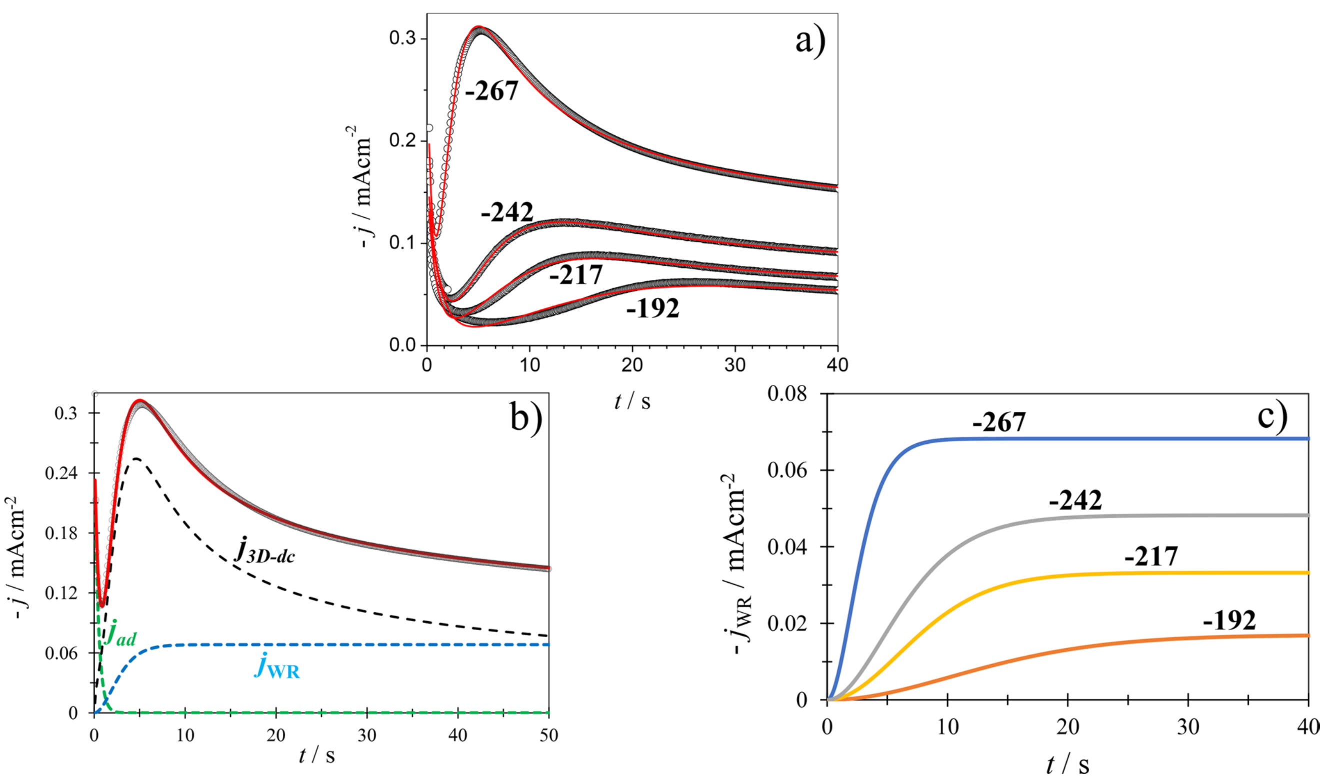

Fig. 3 (**a**) Comparison of theoretical current density transients (solid red lines) obtained after fitting Eq. (14) into the experimental data (points) of the *j-t* plots reported in Fig. 2a. (**b**) Comparison of a theoretical transients (solid red line) obtained after fitting Eq. (14) into the experimental data (points) recorded at η = −267 mV along with the individual contributions (dashed lines) due to $j_{ad}(t)$, associated with adsorption process and doble layer charging, mass-transfer controlled 3D nucleation, $j_{3D}(t)$, and water reduction, $j_{WR}(t)$. The best fit parameters are shown in Table 1

Table 1 Best-fit parameters obtained from the experimental potentiostatic current density transients depicted in Figs. 3a using Eq. (14)

/ mV	k_1	k_2	P_1	P_2 / ms^{-1}	P_3 / $mAcm^{-2}$	A / ms^{-1}
192	160 ± 2	0.85 ± 0.01	760 ± 3	168 ± 1	17 ± 0.2	39 ± 1
217	170 ± 2	1.26 ± 0.02	761 ± 4	194 ± 5	33.2 ± 0.1	67 ± 2
242	176 ± 1	1.02 ± 0.01	762 ± 3	253 ± 1	48.2 ± 0.2	147 ± 3
267	282 ± 3	2.36 ± 0.03	768 ± 7	500 ± 1	68.3 ± 0.1	354 ± 4

with the number density of active sites, N_0, (see Eq. (12), increase exponentially with η. In contrast, P_1, which relates to the diffusion coefficient of Se(IV) in ethaline ($D_{Se(IV)}$), remains essentially constant with overpotential. Based on P_1, (see Eq. (12), the diffusion coefficient was determined as: $D_{Se(IV)}$ (298 K) = (4.91 ± 0.04) × 10^{-9} $cm^2 \cdot s^{-1}$.

In comparison, Bougouma et al. [55]. reported a value of (6.3 ± 0.3) × 10^{-7} $cm^2 \cdot s^{-1}$ for Se(VI) in reline (choline chloride: urea) at 383 K, using potentiostatic *j–t* data and the Cottrell equation.

Furthermore, the overpotential dependence of the nucleation frequency (Fig. 4) enables the estimation of fundamental thermodynamic and kinetic parameters. By fitting Eq. (15) to the data, quantities such as the Gibbs free energy of formation of the critical nucleus ($\Delta G(n_c)$), the number of atoms in the critical nucleus (n_c), the exchange current density (j_0), the charge transfer coefficient (α), and the surface tension (σ) at the electrode/deposit/electrolyte interface were extracted, following the methodology of Sebastián et al. [56].

$$\ln(A) = K_1 + K_2\eta - K_3\eta^{-2} \tag{15}$$

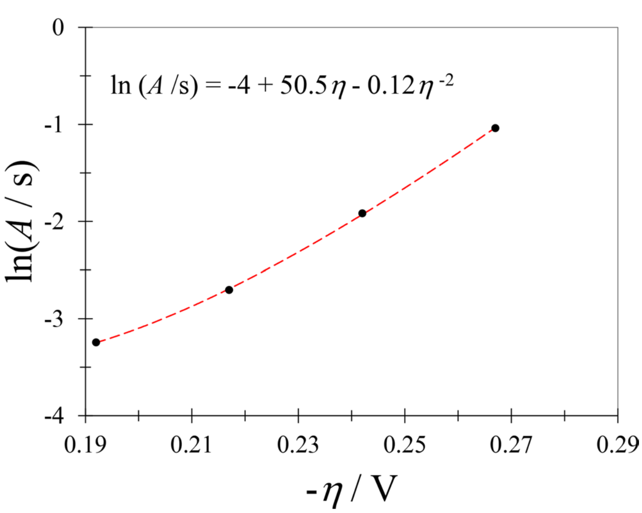

Fig. 4 Overpotential dependence of the nucleation frequency obtained from analysis of experimental density transients shown in Fig. 3a, see also Table 1. The broken line and equation were obtained after non-linear fit of Eq. (15)

with

$$K_1 = ln\left(\frac{\sqrt{2}v_a j_0 \sigma^{1/2}}{z(kT)^{1/2}e}\right) \tag{16}$$

$$K_2 = \frac{z\alpha e}{kT} \tag{17}$$

$$K_3 = -\frac{8\pi\sigma^3 v_a^2}{3(ze)^2 kT} \tag{18}$$

Where k is the Boltzmann constant, e is the elemental charge and v_a is the atomic volume.

These parameters are summarized in Table 2. Notably, the α and j_0 values are consistent with those obtained independently from potentiodynamic studies. Nevertheless, the j_0 value obtained from the overpotential dependence of the nucleation frequency is a more reliable quantity than that obtained from Tafel analysis, since the latter was performed on the reverse scan, where a selenium film is already present and, therefore, the electrode surface area is not strictly constant. The estimated σ and n_c values also agree well with literature data for Ag and Pb nucleation on glassy carbon from aqueous and DES media [56], as well as for Nd from DES [21].

The nucleation work corresponds to the difference between the Gibbs free energy of an aggregate of n atoms in the new phase (on the electrode) and the same number of metal ions in solution. This nucleation work is a function of supersaturation, which in electrochemical systems is related to overpotential and reaches a maximum at the critical size n_c. The corresponding free energy of nucleation, $\Delta G(n_c)$, is an overpotential-dependent quantity described by Eq. (19) [57]. Figure 5 shows the variation of $\Delta G(n_c)$ as a function of η, Fig. 5a, and 1/ η^2, Fig. 5b. As observed, $\Delta G(n_c)$ decreases exponentially with increasing

Table 2 Kinetic and thermodynamic parameters resulting from fitting Eq. (15) to the ln (*A*) vs. η (see Fig. 4) obtained from analysis of the experimental *j-t* plots reported in Fig. 3a

α^{i}	j_0^{ii}	$10^5\sigma^{iii}$ /J cm^{-2}	n_c^{iv}
0.3 ± 0.1	3.5 ± 0.2	3.19 ± 0.02	< 1

Estimated using: *i* (Eq. (17)), *ii* (Eq. (16)), *iii* (Eq. (18)) and *iv* (Eq. (20))

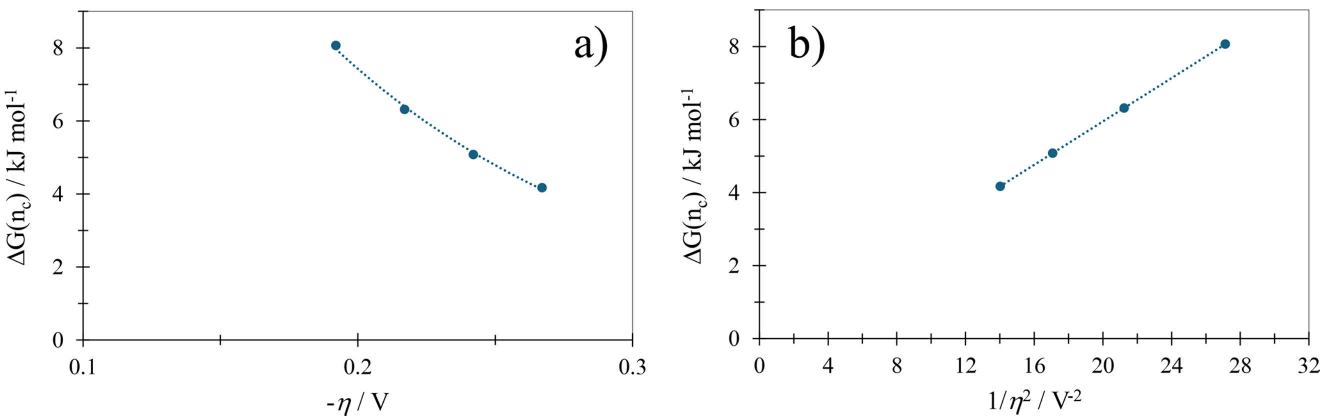

Fig. 5 Variation of $\Delta G(n_c)$, markers, obtained from analysis of ln$A$ vs.η plot, see Fig. 4, using Eq. (15), plotted as a function of (**a**) η and (**b**) $1/\eta^2$, The dotted line represents the exponential (**a**) and linear (**b**) fit to the experimental data respectively

overpotential, indicating a more favorable nucleation process at higher η. In Fig. 5b, $\Delta G(n_c)$ shows a linear dependence on $1/\eta^2$, in agreement with the theoretical prediction of Eq. (19). Remarkably, the critical nucleus size estimated for this system was less than one atom, suggesting that a single active site on the glassy carbon electrode can effectively serve as a critical nucleus [58].

$$\triangle G\left(n_c\right) = \frac{8\pi\sigma^3 v_a^2}{3\left(ze\right)^2\eta^2} = -\frac{kTK_3}{\eta^2} \tag{19}$$

$$n_c = \frac{2\triangle G\left(n_c\right)}{ze\eta} \tag{20}$$

An important aspect of metal electrodeposition is the estimation of Faradaic efficiency (*FE*). This parameter reflects how effectively the cathode utilizes the applied charge during an electrochemical process, specifically the fraction of the total charge, q, or charge density, Q, that is consumed by the desired reduction reaction, in this case, selenium deposition (Q_{3D}), relative to the total charge passed, Q_{total}(Q_{total} = Q_{ad} + Q_{3D}+ Q_{WR}), see Eq. (21)).

In the present work, and owing to the theoretical model employed to analyze the experimental current density–time (j-t) transients (see Eq. (2)), the charge contributions required to evaluate the *FE* for selenium electrodeposition were determined by integrating the individual components of the total current density transient recorded during selenium nucleation and growth (see, for example, Fig. 3b). Figure 6a illustrates, as a representative case, the time evolution of the charge contributions obtained from the j-$t$transient recorded at an overpotential of $\eta = -267$mV, while Fig. 6b compares the temporal evolution of the *FE* for two different applied overpotentials. These results indicate that the Faradaic efficiency for selenium electrodeposition can be modulated by both the deposition time and the applied overpotential. As shown in Fig. 6(b), for both overpotential values considered, the Faradaic efficiency (FE) increases sharply at short times and reaches a maximum. Therefore, this behavior must be taken into account to ensure efficient fabrication of very thin films or ultra-small nanoparticles.

$$FE\,(\%) = 100\left(\frac{Q_{3D}}{Q_{Total}}\right) \tag{21}$$

Interestingly, Hu et al. [59] have developed high-fidelity bioelectrodes that integrate multiple charge-transfer processes, like those observed in the present work during Se electrodeposition. Therefore, the use of Se nanoparticles (SeNPs) in biosensing applications may contribute to advances in bioelectrode design, particularly those requiring bidirectional ion–electron transduction for such applications.

SEM, EDX and XPS analysis

SEM images recorded on the GCE surface after selenium potentiostatic deposition (Fig. 7a and c) reveal the formation of uniformly distributed quasi-spherical and smooth selenium nanoparticles (SeNPs), as highlighted in EDX spectrum reported in Fig. 7d. These SeNPs exhibit an average diameter of (118±26) nm, as determined from size distribution analysis shown in Fig. 7e. Furthermore, XPS analysis confirmed the presence of elemental selenium (Se^0) on the electrode surface, as shown in Fig. 8. These findings are consistent with XPS spectra reported by Bisht et al. [1] and Shenasa et al. [60]. It is worth mentioning that Ye et al. [61] reported the electrochemical formation of amorphous SeNPs from aqueous media.

Raman and UV–Vis analyses

The Raman spectrum reported in Fig. 9a depicts the two characteristic Raman bands at 236 cm^{-1} and 141 cm^{-1} corresponding to trigonal-Se [62] and those at 1600 cm^{-1} and

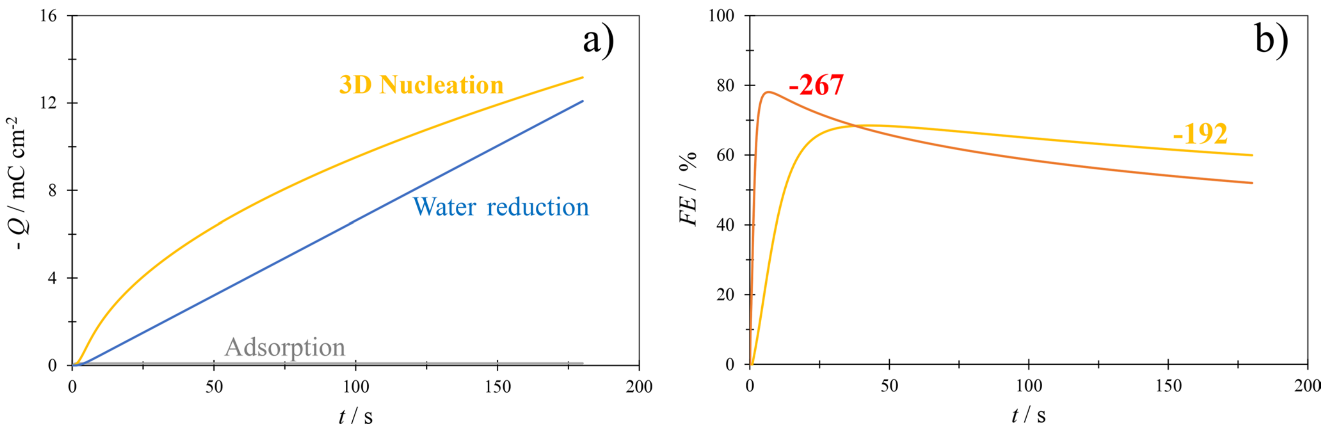

Fig. 6 (**a**) Time variation of the charge density, Q, obtained by integration of the different contributions, to the total current density, recorded during Se electrodeposition at $\eta = -267$ mV, see Fig. 3b. (**b**) Selenium electrodeposition Faradaic Efficiency (FE) as a function of time for the different applied overpotential indicated in the figure in mV

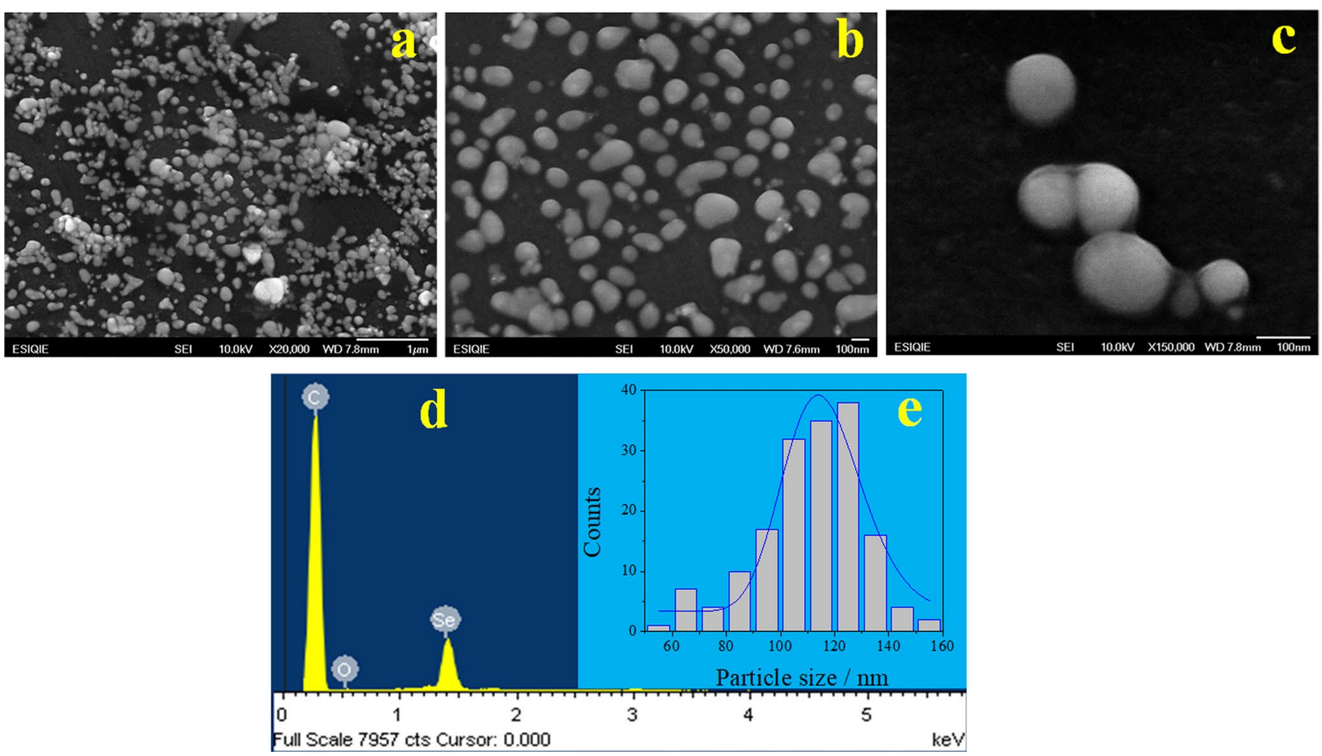

Fig. 7 SEM images of the GCE surface after Se electrodeposition at an overpotential of −267 mV for 50 s (see Fig. 3b), recorded at different magnifications (**a**) 20,000X, (**b**) 50,000X and (**c**) 150,000X. (**d**) EDS spectrum recorded on the surface shown in a). (**e**)Particle size distribution derived from the SEM images

1350 cm^{-1} associated with the carbonaceous substrate [63], It is worth mentioning that the Raman spectra were acquired using a 532 nm green laser operated at 50% of its maximum power, providing a penetration depth of several micrometers into the material. Measurements were performed at multiple regions of each sample to confirm the homogeneity of the Raman response and to ensure that the recorded spectra were representative of the entire surface.

The relationship between the size of selenium nanoparticles and their Raman intensity is a significant effect, with smaller nanoparticles exhibiting higher Raman signal intensity, while carbon-related features appear as a secondary contribution in the spectrum. This behavior arises from the relatively larger contribution of small Se nanoparticles compared to bulk material, as the measured signal reflects the collective response of the nanostructures. Consequently,

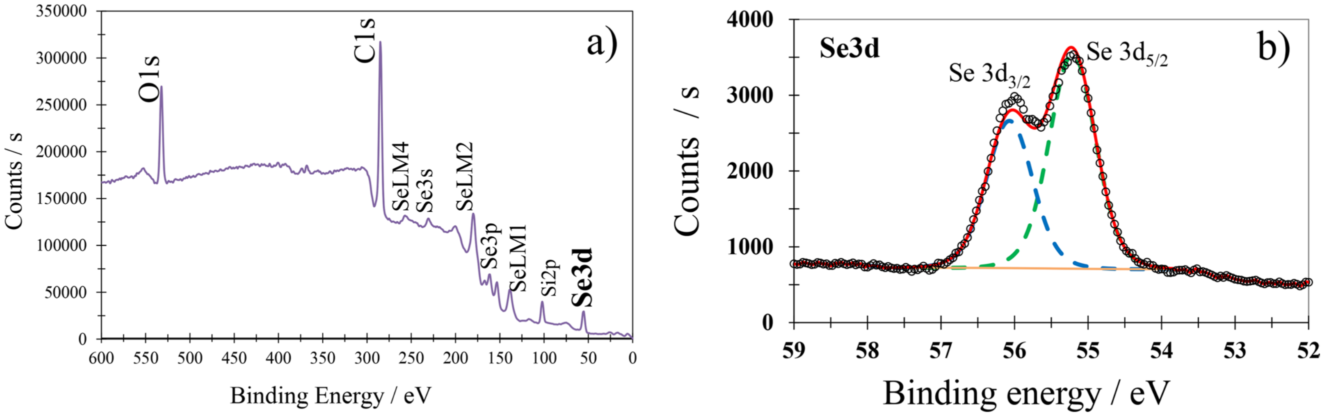

Fig. 8 (**a**) Survey XPS spectrum recorded on the surfaces shown in Fig. 6a (**b**) High-resolution XPS spectrum of the Se 3d region, including both $3d_{5/2}$ and $3d_{3/2}$ components

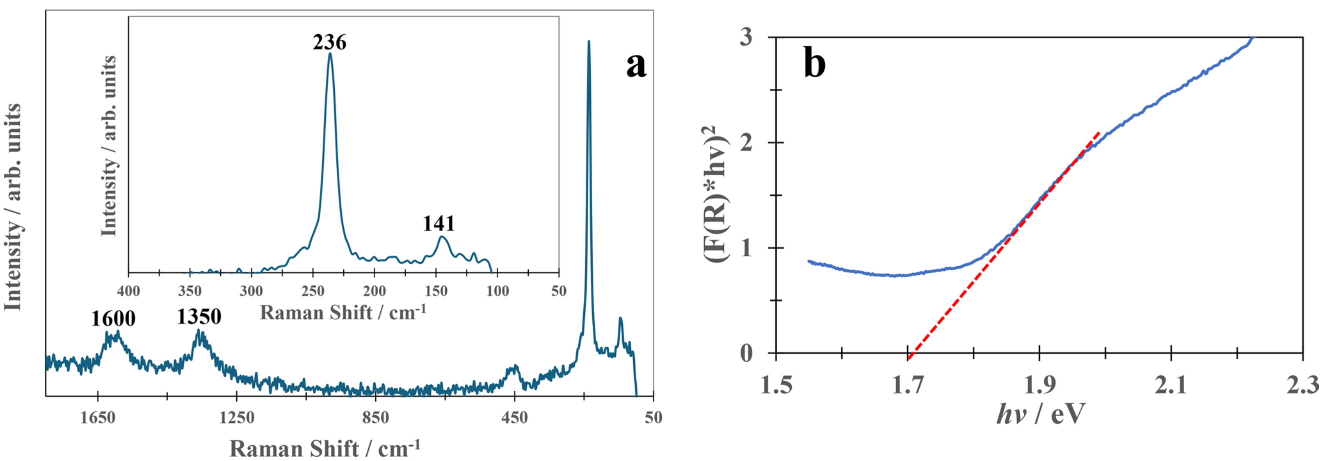

Fig. 9 (**a**) Raman spectrum and (**b**) Kubelka-Munk transform, $(F(R)*hv)^2$ vs. *hv* corresponding to the surface shown in Fig. 6a recorded at 298 K

regions with a higher density of Se nanostructures exhibit enhanced Raman intensity, resulting in a stronger and more representative nanoparticle signal. Notably, Saha et al. [64] also reported the electrodepostion of trigonal-Se onto GCE at 298 and 323 K from an amide type hydrophobic room temperature ionic liquid, 1-butyl-1-methylpyrrolidinium bis(trifluoromethylsulfonyl)amide (BMPTFSA) containing $SeCl_4$ as a Se source in the presence of excess chloride, meanwhile Seyedmahmoudbaraghani et al. [65] reported the electrodepostion of amorphous Se from aqueous media at pH 1.5 and 353 K.

To further verify the formation of trigonal SeNPs, UV–Vis absorption spectrum was recorded (it is worth mentioning that, prior to the UV–Vis diffuse reflectance spectroscopy measurements, the instrument was calibrated using a Spectralon reference sample (Labsphere). Spectralon is a sintered polytetrafluoroethylene (PTFE) fluoropolymer, making it an ideal reference standard for optical measurements. Its diffuse white appearance arises from strong bulk scattering properties, which enable accurate calibration. Such standards, commonly employed as diffuse reflectance targets, allow precise characterization of instrument linearity and spectral response across the UV, visible, and near-infrared wavelength ranges) and the corresponding Kubelka-Munk transform, $(F(R)*hv)^2$ vs. *hv*. plot was constructed (see Fig. 9b). The obtained spectra exhibit a pronounced absorption edge attributable to the electronic band gap, together with several secondary, weakly defined edge-like features that can be mainly associated with indirect band-gap transitions. This electronic structure enables a detailed characterization of the band-gap magnitude and its direct or indirect nature, while also providing insight into the presence of localized and delocalized electronic states, the effective mass of charge carriers, and other fundamental electronic properties.

According to the calculated electronic band structure, and as observed in many other semiconductor systems, the band gap and other electronic transitions of selenium nanoparticles exhibit a blue shift with decreasing nanoparticle size. The band-gap energy was determined as the intersection point with the x-axis, obtained by extrapolating the tangent drawn at the inflection point of the plotted curve to the photon energy [hv] axis. From this plot the optical band gap was calculated to be 1.7 eV, which is consistent with values reported for trigonal Se nanowires [66].

Conclusions

Successful Electrodeposition of Trigonal SeNPs. Zero-valent, trigonal selenium nanoparticles (SeNPs) were successfully synthesized via potentiostatic electrodeposition from SeO_2 dissolved in ethaline at room temperature. This confirms the suitability of deep eutectic solvents (DESs) as green and efficient media for controlled SeNP fabrication.

Mechanistic and Kinetic Insights. The electrochemical nucleation and growth of SeNPs on glassy carbon electrodes follow a complex mechanism involving double-layer charging, multiple 3D nucleation, and diffusion-controlled growth, along with parasitic hydrogen evolution. These findings were quantitatively supported by kinetic parameters extracted from deconvoluted *j–t* transients.

Thermodynamic and Electrochemical Characterization. The estimated thermodynamic and electrochemical parameters, such as the equilibrium potential (E_{eq} = 96 mV), exchange current density (j_0 = (1.2 ± 0.1) μA·cm^{-2}), charge transfer coefficient ($\alpha = 0.4 \pm 0.1$), diffusion coefficient ($D_{Se(IV)} = 4.91 \times 10^{-9}$ cm^2·s^{-1}), surface energy, and free energy of nucleus formation, enhance our understanding of selenium electrocrystallization in DESs.

Morphological and Structural Validation. SEM and XPS analyses confirmed the formation of quasi-spherical SeNPs (~ 118 ± 26 nm) composed of elemental selenium (Se^0), while Raman spectroscopy validated the trigonal (t-Se) allotropic form as the predominant phase.

Optical Properties Consistent with t-Se. UV–Vis spectroscopy revealed an optical band gap of 1.7 eV, in agreement with previously reported values for t-Se, indicating the semiconductor nature of the synthesized SeNPs and their potential applicability in optoelectronic and sensing devices.

Overall, this study introduces a novel, one-step method for synthesizing selenium nanoparticles via electrochemical deposition from a DES medium. It not only offers a greener and more controllable alternative to conventional synthesis methods but also fills a significant gap in the literature by elucidating, for the first time, the nucleation and growth mechanism of SeNPs in this system, laying the groundwork for their optimized use in sensing, biomedical, and energy applications.

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Data availability Data will be made available on request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
The manuscript was entirely human written.

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